

Solutions: Additional Exercises 3

Power Series

Series and Integral Transforms

Exercise 1 — Radius and Domain of Convergence

Method: find radius R via Ratio or Root test, then check each endpoint separately.

(a) ★ $\sum_{n=1}^{\infty} \frac{x^n}{3^n \cdot n}$

Coefficients $a_n = 1/(3^n n)$. By the Ratio Test: $R = \lim \frac{|a_n|}{|a_{n+1}|} = \lim \frac{3^{n+1}(n+1)}{3^n n} = 3$.

Endpoints:

- $x = 3$: $\sum 1/n$ (harmonic) — **diverges**.
- $x = -3$: $\sum (-1)^n/n$ — Leibniz, **converges**.

Domain: $[-3, 3)$.

(b) ★ $\sum_{n=0}^{\infty} \frac{x^n}{n!}$

Ratio Test: $\lim \frac{n!}{(n+1)!} |x| = \lim \frac{|x|}{n+1} = 0$ for all x . So $R = +\infty$.

Domain: $(-\infty, +\infty)$.

(c) ★ $\sum_{n=1}^{\infty} \frac{(-1)^n x^n}{n}$

$R = 1$ (Ratio Test).

Endpoints:

- $x = 1$: $\sum (-1)^n/n$ — Leibniz, **converges**.
- $x = -1$: $\sum (-1)^n(-1)^n/n = \sum 1/n$ — **diverges**.

Domain: $(-1, 1]$.

(d) ★ $\sum_{n=0}^{\infty} n! x^n$

Ratio Test: $\lim(n+1)|x| = +\infty$ for $x \neq 0$. So $R = 0$.

Domain: $\{0\}$.

(e) ★★ $\sum_{n=1}^{\infty} \frac{(-1)^n x^{2n-1}}{5^n \sqrt{n+1}}$

The series has only odd powers of x . Write $u = x^2$. The coefficients (of u^n) satisfy:

$$R_u = \lim \frac{5^{n+1} \sqrt{n+2}}{5^n \sqrt{n+1}} = 5.$$

So $|x^2| < 5$, meaning $|x| < \sqrt{5}$.

Endpoints $x = \pm\sqrt{5}$: the series becomes $\sum \pm 1/\sqrt{n+1}$, which diverges ($p = 1/2$).

Domain: $(-\sqrt{5}, \sqrt{5})$.

(f) ★★ $\sum_{n=0}^{\infty} \frac{2^n x^n}{(2n+1)^2 n^{1/4}}$

Ratio Test gives $R = 1/2$.

Endpoints $x = \pm 1/2$: the series becomes $\sum \frac{\pm 1}{(2n+1)^2 n^{1/4}}$. Since $(2n+1)^2 n^{1/4} \sim 4n^{9/4}$ and $p = 9/4 > 1$, this **converges absolutely**.

Domain: $[-\frac{1}{2}, \frac{1}{2}]$.

(g) ★★ $\sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^{2n-2}}{3^{n-1} \cdot n \sqrt{n}}$

Set $u = x^2$; $R_u = 3$. So $|x| < \sqrt{3}$.

Endpoints $x = \pm\sqrt{3}$: $\sum \frac{1}{n\sqrt{n}} = \sum n^{-3/2}$ converges ($p = 3/2$).

Domain: $[-\sqrt{3}, \sqrt{3}]$.

(h) ★★★★★ $\sum_{n=0}^{\infty} \frac{(-1)^n \ln(n+2)(2x+3)^n}{(n+1)(2n)!}$

The coefficients involve $(2n)!$ in the denominator, so the Ratio Test gives $R = +\infty$ for the variable $u = 2x + 3$. Since this holds for all $u \in \mathbb{R}$, the series converges for all x .

Domain: $(-\infty, +\infty)$.

(i) ★★★★★ $\sum_{n=1}^{\infty} \frac{1 \cdot 3 \cdots (2n-1)}{2 \cdot 4 \cdots (2n)} x^n$

Ratio Test: $\frac{a_{n+1}}{a_n} = \frac{2n+1}{2n+2} \rightarrow 1$. So $R = 1$.

Endpoints: by Wallis, $a_n \sim 1/\sqrt{\pi n}$, so $\sum a_n$ diverges at both $x = \pm 1$.

Domain: $(-1, 1)$.

$$(j) \quad \star\star\star\star \sum_{n=0}^{\infty} \frac{(2n)! x^n}{4^n (n!)^2}$$

Same coefficients as above (recognising the central binomial). Ratio Test gives $R = 1$; endpoints diverge.

Domain: $(-1, 1)$.

$$(k) \quad \star\star \sum_{n=1}^{\infty} \frac{(3n-1)(-1)^n x^n}{n^3}$$

$$R = \lim_{n \rightarrow \infty} \frac{|a_n|}{|a_{n+1}|} = \lim_{n \rightarrow \infty} \frac{(3n-1)}{n^3} \cdot \frac{(n+1)^3}{3n+2} = 1.$$

Endpoints:

- $x = 1$: $\sum (3n-1)(-1)^n/n^3$. Since $(3n-1)/n^3 \searrow 0$, **Leibniz** applies; converges.
- $x = -1$: $\sum (3n-1)/n^3 \sim \sum 3/n^2$; converges ($p = 2$).

Domain: $[-1, 1]$.

Exercise 2 — Taylor Series around a Point a

(a) e^x around $a = 0$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad R = \infty.$$

(b) $\sin x$ around $a = 0$

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \quad R = \infty.$$

(c) $\frac{1}{1-x}$ around $a = 0$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n, \quad |x| < 1.$$

(d) $\star\star \frac{1}{x-5}$ around $a = 1$

Write $x-5 = (x-1)-4$, so:

$$\frac{1}{x-5} = \frac{1}{(x-1)-4} = -\frac{1}{4} \cdot \frac{1}{1-\frac{x-1}{4}} = -\sum_{n=0}^{\infty} \frac{(x-1)^n}{4^{n+1}}, \quad |x-1| < 4.$$

Domain: $(-3, 5)$.

(e) ★★ $\sin x$ **around** $a = \pi/2$

Let $t = x - \pi/2$. Then $\sin x = \sin(t + \pi/2) = \cos t = \sum_{n=0}^{\infty} \frac{(-1)^n t^{2n}}{(2n)!}$:

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n (x - \pi/2)^{2n}}{(2n)!}, \quad R = \infty.$$

(f) ★ $\ln x$ **around** $a = 1$

$$\ln x = \ln(1 + (x - 1)) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} (x - 1)^n}{n}, \quad 0 < x \leq 2.$$

(g) ★★★ $\frac{1}{(x - 5)^3}$ **around** $a = 1$

From part (d): $\frac{1}{x - 5} = -\sum_{n=0}^{\infty} \frac{(x - 1)^n}{4^{n+1}}$.

Differentiate twice with respect to x :

$$\frac{1}{(x - 5)^2} = \sum_{n=1}^{\infty} \frac{n(x - 1)^{n-1}}{4^{n+1}}, \quad \frac{2}{(x - 5)^3} = \sum_{n=2}^{\infty} \frac{n(n - 1)(x - 1)^{n-2}}{4^{n+1}}.$$

Dividing by 2 and re-indexing ($m = n - 2$):

$$\frac{1}{(x - 5)^3} = \sum_{m=0}^{\infty} \frac{(m + 2)(m + 1)}{2 \cdot 4^{m+3}} (x - 1)^m, \quad |x - 1| < 4.$$

(h) ★★ $\frac{1}{(x - 2)^2}$ **around** $a = -3$

Write $x - 2 = (x + 3) - 5$, so $\frac{1}{x - 2} = -\frac{1}{5} \cdot \frac{1}{1 - (x + 3)/5} = -\sum_{n=0}^{\infty} \frac{(x + 3)^n}{5^{n+1}}$.

Differentiate with respect to x :

$$\frac{1}{(x - 2)^2} = \sum_{n=1}^{\infty} \frac{n(x + 3)^{n-1}}{5^{n+1}}, \quad |x + 3| < 5.$$

Domain: $(-8, 2)$.

Exercise 3 — Maclaurin Series (Substitution)

(a) ★ e^{-x^2}

Substitute $u = -x^2$ into $e^u = \sum u^n/n!$:

$$e^{-x^2} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{n!}, \quad \forall x \in \mathbb{R}.$$

(b) ★ $\sin(x^3)$

Substitute $u = x^3$ into $\sin u = \sum (-1)^n u^{2n+1}/(2n+1)!$:

$$\sin(x^3) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{6n+3}}{(2n+1)!}, \quad \forall x \in \mathbb{R}.$$

(c) ★★ $\frac{1}{x^2 + 4x + 5}$

Complete the square: $x^2 + 4x + 5 = (x+2)^2 + 1$. Let $u = -(x+2)^2$:

$$\frac{1}{(x+2)^2 + 1} = \frac{1}{1-u} \Big|_{u=-(x+2)^2} = \sum_{n=0}^{\infty} (-1)^n (x+2)^{2n}.$$

This is a Taylor series centered at $a = -2$, not $a = 0$. For a Maclaurin series ($a = 0$), write $v = (4x + x^2)/5$ so $x^2 + 4x + 5 = 5(1+v)$:

$$\frac{1}{5} \cdot \frac{1}{1+v} = \frac{1}{5} \sum_{n=0}^{\infty} (-1)^n v^n = \frac{1}{5} \sum_{n=0}^{\infty} (-1)^n \left(\frac{x^2 + 4x}{5} \right)^n, \quad |x^2 + 4x| < 5.$$

(d) ★ xe^x

Multiply $e^x = \sum x^n/n!$ by x :

$$xe^x = \sum_{n=0}^{\infty} \frac{x^{n+1}}{n!} = \sum_{n=1}^{\infty} \frac{x^n}{(n-1)!}, \quad \forall x.$$

(e) ★ $\ln(1+x^2)$

Substitute $u = x^2$ into $\ln(1+u) = \sum_{n=1}^{\infty} (-1)^{n+1} u^n/n$:

$$\ln(1+x^2) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^{2n}}{n}, \quad |x| \leq 1.$$

(f) ★★★ $\arctan\left(\frac{1+x^2}{1-x^2}\right)$

Use the tangent addition identity: $\frac{1+x^2}{1-x^2} = \tan\left(\frac{\pi}{4} + \arctan(x^2)\right)$ (verify by $\tan A + \tan B$ formula with $A = \pi/4$, $\tan A = 1$, $\tan B = x^2$, $\tan(A+B) = (1+x^2)/(1-x^2)$).

Therefore $\arctan\left(\frac{1+x^2}{1-x^2}\right) = \frac{\pi}{4} + \arctan(x^2)$.

Using $\arctan u = \sum_{n=0}^{\infty} \frac{(-1)^n u^{2n+1}}{2n+1}$ and substituting $u = x^2$:

$$f(x) = \frac{\pi}{4} + \sum_{n=0}^{\infty} \frac{(-1)^n x^{4n+2}}{2n+1}, \quad |x| < 1.$$

Exercise 4 — High-Order Derivatives via Taylor Coefficients

(a) ★★ $f^{(17)}(1)$ for $f(x) = \frac{1}{x-5}$

From Exercise 2(d): $\frac{1}{x-5} = -\sum_{n=0}^{\infty} \frac{(x-1)^n}{4^{n+1}}$.

The Taylor coefficient of $(x-1)^{17}$ is $-1/4^{18}$. Since $c_{17} = f^{(17)}(1)/17!$:

$$f^{(17)}(1) = 17! \cdot \left(-\frac{1}{4^{18}}\right) = \boxed{-\frac{17!}{4^{18}}}.$$

(b) ★★ $\frac{d^{10}}{dx^{10}} \left(\frac{1}{1-x} \right) \Big|_{x=0}$

The Maclaurin series is $\sum_{n=0}^{\infty} x^n$. The coefficient of x^{10} is 1. Since $c_{10} = f^{(10)}(0)/10!$:

$$f^{(10)}(0) = 10! \cdot 1 = \boxed{10!} = 3\,628\,800.$$

(c) ★★★ **Evaluating** $\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (1-a)^n$

This expression is precisely the Taylor series of f centered at a , evaluated at $x = 1$:

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (1-a)^n = f(1),$$

provided $|1-a| < R_a$ (the radius of convergence of the Taylor series at a). Since by hypothesis $R_a > 1$, we have $|1-a| = |1-a| \leq 1 < R_a$, so the series converges to $f(1)$. $\square$

Key takeaways: The radius R is found by Ratio/Root test on coefficients. Endpoint analysis requires separate care. Substitution is the fastest way to build Maclaurin series for composite functions. Taylor coefficients directly give all derivatives at the center: $f^{(n)}(a) = n! \cdot c_n$.